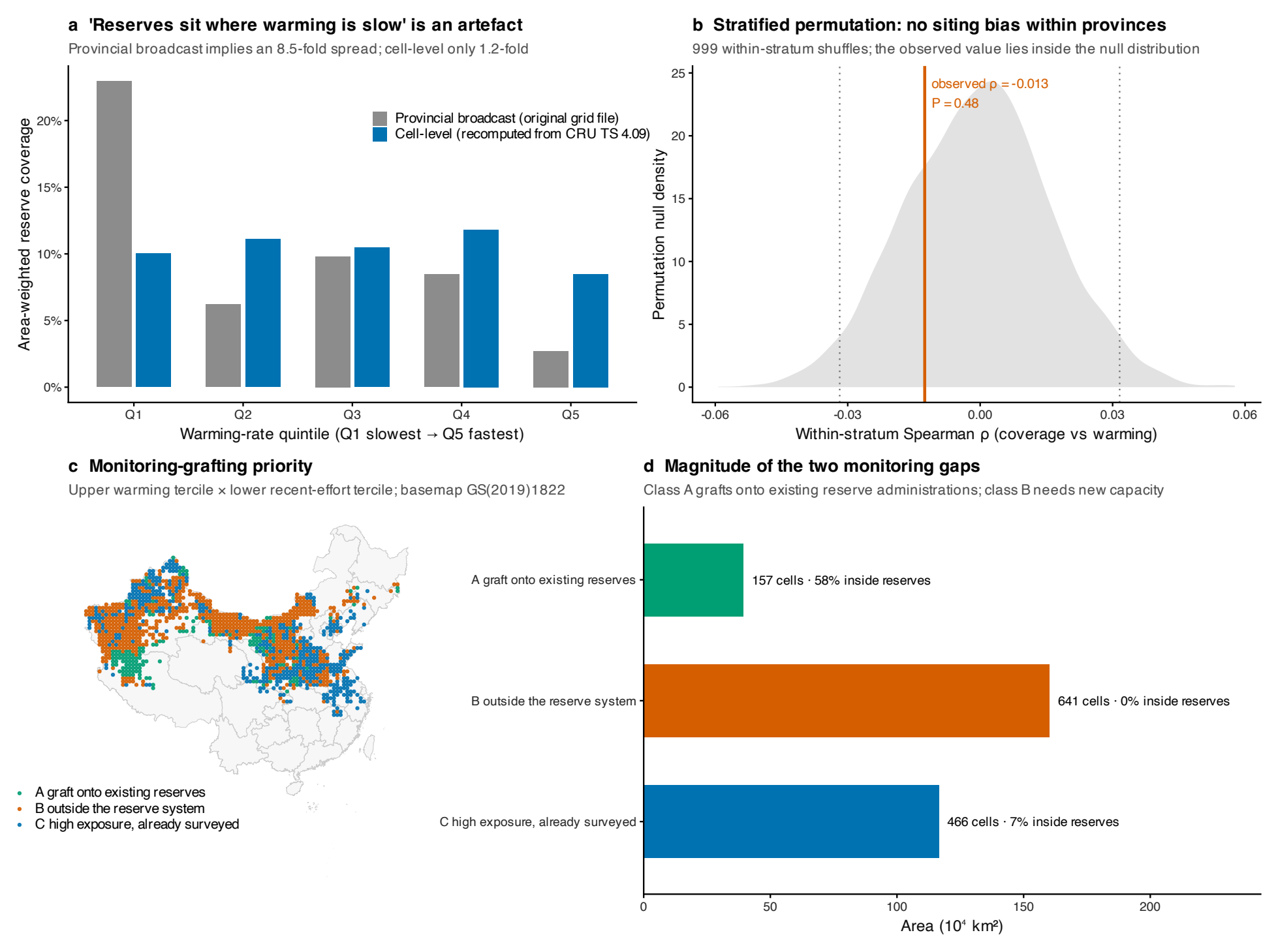

a 'Reserves sit where warming is slow' is an artefact
Provincial broadcast implies an 8.5-fold spread; cell-level only 1.2-fold
Provincial broadcast (original grid file)
Cell-level (recomputed from CRU TS 4.09)
Area-weighted reserve coverage
20%
15%
10%
5%
0%
Q1
Q2
Q3
Q4
Q5
Warming-rate quintile (Q1 slowest → Q5 fastest)
b Stratified permutation: no siting bias within provinces
999 within-stratum shuffles; the observed value lies inside the null distribution
observed ρ = -0.013
P = 0.48
Permutation null density
25
20
15
10
5
0
-0.06
-0.03
0.00
0.03
0.06
Within-stratum Spearman ρ (coverage vs warming)
c Monitoring-grafting priority
Upper warming tercile × lower recent-effort tercile; basemap GS(2019)1822
A graft onto existing reserves
B outside the reserve system
C high exposure, already surveyed
d Magnitude of the two monitoring gaps
Class A grafts onto existing reserve administrations; class B needs new capacity
A graft onto existing reserves
157 cells · 58% inside reserves
B outside the reserve system
641 cells · 0% inside reserves
C high exposure, already surveyed
466 cells · 7% inside reserves
0
50
100
150
200
Area (10⁴ km²)